INCH-POUND
MIL-B-85667/9
28 January 1991

MILITARY SPECIFICATION SHEET

BOLT, SLEEVE, PROTRUDING HEAD, STRAIGHT SHANK, CRES 304 SLEEVE, CRES A-286 PIN, 95 KSI Fsu, 160 KSI Ftu, 450°F

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-85667.

1. Application criteria. The sleeve bolt shall be used to provide a precision net fit or controlled interference fit. The sleeve bolts shall be used in accordance with the limitations of MIL-B-85667/1(AS). The sleeve bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-85729 shall be used with these sleeve bolts.

3. Oversize bolts. First and second oversize sleeve bolts may replace MIL-B-85667/9 sleeve bolts for first and second repair operations.

4. Installation. Installation shall be in accordance with MIL-HDBK-271-104.

5. Materials.

5.1 Pin. Pin material shall be Cres steel A-286 in accordance with AMS 5732 or AMS 5737 (UNS S66286).

5.2 Cold work. Pin shall be cold worked and aged to 160 KSI Ftu and 95 KSI Fsu.

5.3 Sleeve. Sleeve material shall be corrosion resistant steel, AISI 304 (UNS S30400), in accordance with MIL-T-8504.

6. Heat treatment.

6.1 Pin. The pin shall be heat treated in accordance with AMS 5732 or AMS 5737.

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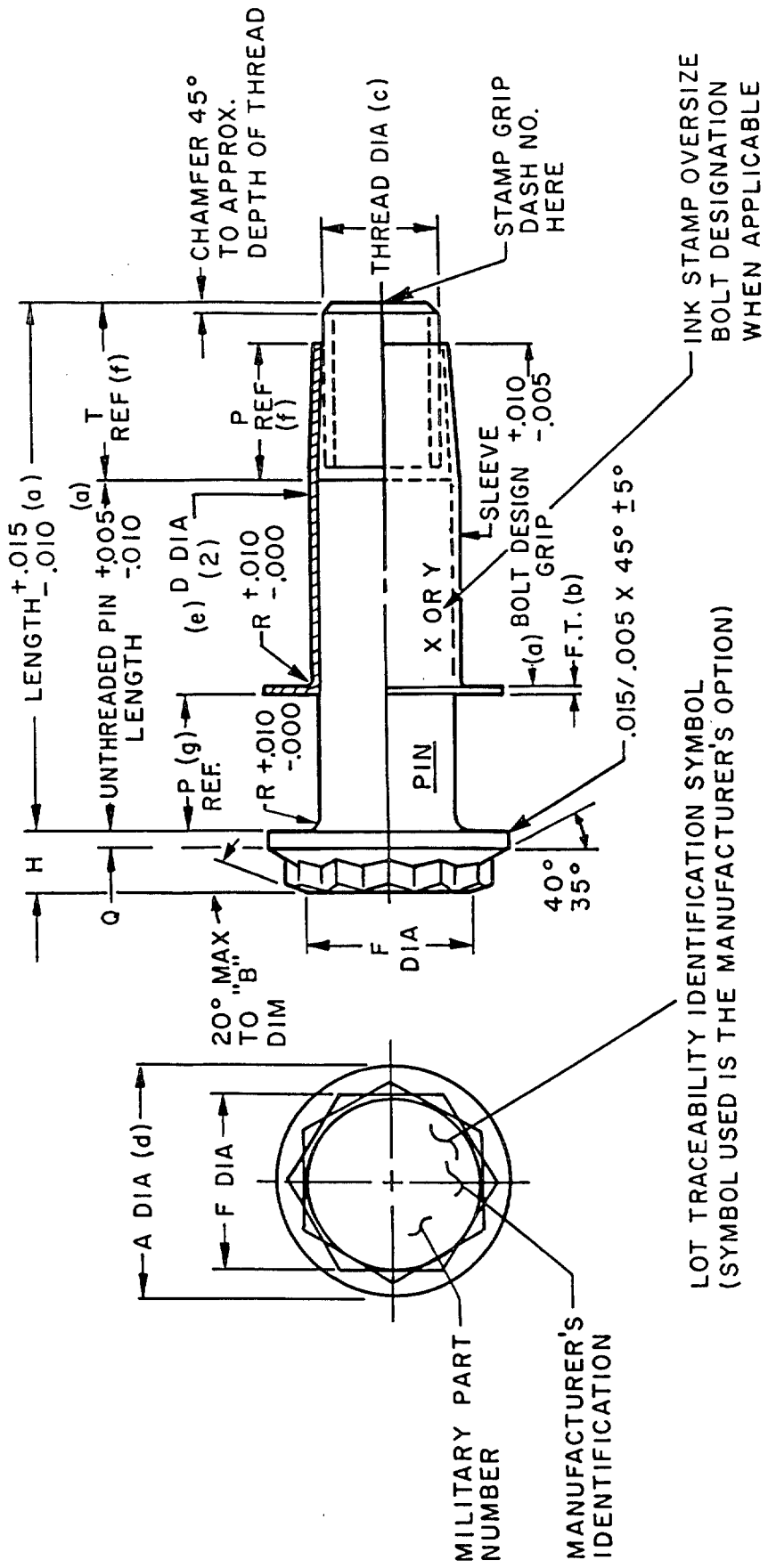


FIGURE 1. Flanged tension head bolt.

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NOTES:

a. Grip lengths are available in .0625 increments.

b. Dimension of non-structural flange thickness is established by sleeve bolt assembly installation requirements allowing no threads in bearing.

c. Threads shall be in accordance with MIL-S-8879. Thread major diameter (T.D.) shall be in accordance with Table I.

d. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 FIM.

2. For sizes .437 thru .625, .012 FIM.

3. For sizes .750 thru 1.000, .020 FIM.

e. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 FIM.

2. For sizes .437 thru 1.000, .010 FIM.

f. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

g. Protrusion will vary according to taper to provide expansion as given in Table I.

6.1.1 Hardness. Hardness shall not be greater than 45 HRC.

6.2 Sleeve. Sleeve is cold worked.

7. Surface texture. Surface texture shall be R_a-32 in accordance with ANSI/ASME B46.1.

8. Coatings or surface treatment.

8.1 Pin. The pin shall be dry-film lubricated in accordance with MIL-L-46010, Type I, .0003/.0005 thick.

8.2 Sleeve. None.

9. Lubricants. 0.0001-inch overspray allowed on head and threads. After application of dry lubrication, the pin outside diameter and sleeve inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-87132.

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TABLE I. Bolt dimensions.

SIZE NO's	GROUP	GRIP DASH NUMBERS	THREAD PER MIL-S-8879 UNJF-3A	A DIA.	H HEAD HEIGHT	F	F.T. +.015(b) -.000	Q	P (f) REF.	R FILLET RADIUS	T.D. THREAD DIA.	T THREAD LENGTH REF.	D DIMENSION (a) (b) UNINSTALLED SLEEVE DIAMETER			D' DIMENSION (a) (b) EXPANDED SLEEVE DIAMETER			MINIMUM INSTALLATION HOLE SIZE DIAMETER		
													STD.	X O/S	Y O/S	STD.	X O/S	Y O/S	STD.	X O/S	Y O/S
-3	1	-5 THRU -8	.1900-32	.409 .399	.128 .118	.313 .305	.010	.030 .020	.175 .225 .215	.025	.1840 .1810	.343	.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	2	-9 THRU -16							.432 .422				.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	3	-17 THRU -22							.432 .422				.2485 .2480	.2555 .2550	.2625 .2620	.2575 .2565	.2645 .2635	.2715 .2705	.249	.256	.263
-4	1	-5 THRU -8	.2500-28	.516 .506	.138 .126	.376 .367	.010	.045 .035	.175 .225 .215	.025	.2440 .2410	.406	.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	2	-9 THRU -17							.432 .422				.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	3	-18 THRU -28							.432 .422				.3145 .3140	.3215 .3210	.3305 .3300	.3235 .3225	.3305 .3295	.3395 .3385	.315	.322	.331
-5	1	-6 THRU -8	.3125-24	.636 .624	.173 .161	.502 .492	.015	.050 .040	.225 .215	.030	.3060 .3020	.484	.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	2	-9 THRU -20							.432 .422				.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	3	-21 THRU -31							.432 .422				.3805 .3800	.3885 .3880	.3955 .3950	.3895 .3885	.3975 .3965	.4045 .4035	.381	.389	.396
-6	1	-6 THRU -8	.3750-24	.718 .704	.187 .175	.564 .553	.015	.053 .043	.225 .215	.030	.3680 .3640	.562	.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	2	-9 THRU -22							.432 .422				.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	3	-23 THRU -34							.432 .422				.4475 .4470	.4555 .4550	.4635 .4630	.4565 .4555	.4645 .4635	.4725 .4715	.448	.456	.464
-7	1	-6 THRU -8	.4375-20	.808 .792	.218 .206	.627 .616	.020	.056 .046	.225 .215	.040	.4310 .4260	.625	.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	2	-9 THRU -26							.432 .422				.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	3	-27 THRU -40							.432 .422				.5185 .5180	.5295 .5290	.5415 .5410	.5275 .5265	.5385 .5375	.5505 .5495	.519	.530	.542
-8	1	-6 THRU -8	.5000-20	.893 .877	.252 .240	.690 .679	.020	.060 .050	.225 .215	.040	.4930 .4880	.703	.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	2	-9 THRU -30							.432 .422				.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	3	-31 THRU -46							.432 .422				.5895 .5890	.5995 .5990	.6085 .6080	.5985 .5975	.6085 .6075	.6175 .6165	.590	.600	.609

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TABLE I. Bolt dimensions (continued).

SIZE NO's	GROUP	GRIP DASH NUMBERS	THREAD PER MIL-S-8879 UNJF-3A	A DIA.	H HEAD HEIGHT	F	F.T. +.015(b) -.000	Q	P (F) REF.	R FILLET RADIUS	T.D. THREAD DIA.	T THREAD LENGTH REF.	D DIMENSION (a) (b) UNINSTALLED SLEEVE DIAMETER			D' DIMENSION (a) (b) EXPANDED SLEEVE DIAMETER			MINIMUM INSTALLATION HOLE SIZE DIAMETER		
													STD.	X O/S	Y O/S	STD.	X O/S	Y O/S	STD.	X O/S	Y O/S
-9	1	-10 THRU -23	.5625-18	1.096 1.079	.280 .268	.877 .865	.025	.070 .055	.432 .422	.073	.555 .550	.781	.6395 .6390	.6545 .6540	.6675 .6670	.6485 .6475	.6635 .6625	.6765 .6755	.640	.655	.668
	2	-24 THRU -48											.6705 .6700	.6865 .6860	.7015 .7010	.6795 .6785	.6955 .6945	.7105 .7095	.671	.687	.702
	3	-49 THRU -72											.7015 .7010	.7175 .7170	.7325 .7320	.7105 .7095	.7265 .7255	.7415 .7405	.702	.718	.733
-10	1	-10 THRU -25	.6250-18	1.191 1.175	.324 .312	.940 .928	.025	.075 .060	.432 .422	.073	.618 .612	.843	.7075 .7070	.7265 .7260	.7465 .7460	.7165 .7155	.7355 .7345	.7555 .7545	.708	.727	.747
	2	-26 THRU -52											.7465 .7460	.7645 .7640	.7795 .7790	.7555 .7545	.7735 .7725	.7885 .7875	.747	.765	.780
	3	-53 THRU -78											.7795 .7790	.7955 .7950	.8115 .8110	.7885 .7875	.8045 .8035	.8205 .8195	.780	.796	.812
-12	1	-10 THRU -29	.7500-16	1.386 1.370	.392 .378	1.064 1.052	.030	.080 .065	.432 .422	.073	.743 .737	1.000	.8425 .8420	.8575 .8570	.8735 .8730	.8515 .8505	.8665 .8655	.8825 .8815	.843	.858	.874
	2	-30 THRU -60											.8845 .8840	.9045 .9040	.9205 .9200	.8935 .8925	.9135 .9125	.9295 .9285	.885	.905	.921
	3	-61 THRU -90											.9205 .9200	.9365 .9360	.9515 .9510	.9295 .9285	.9455 .9445	.9605 .9595	.921	.937	.952
-14	1	-15 THRU -26	.8750-14	1.589 1.573	.470 .454	1.252 1.239	.030	.090 .075	.432 .422	.073	.868 .861	1.156	.9825 .9820	.9985 .9980	1.0145 1.0140	.9915 .9905	1.0075 1.0065	1.0235 1.0225	.983	.999	1.015
	2	-27 THRU -67					.030		.900 .890				.9825 .9820	.9985 .9980	1.0145 1.0140	.9915 .9905	1.0075 1.0065	1.0235 1.0225	.983	.999	1.015
	3	-68 THRU -108					.035		.900 .890				1.0455 1.0450	1.0615 1.0610	1.0765 1.0760	1.0545 1.0535	1.0705 1.0695	1.0855 1.0845	1.046	1.062	1.077
-16	1	-15 THRU -26	1.0000-12	1.788 1.722	.538 .522	1.440 1.427	.031	.100 .085	.432 .422	.073	.993 .986	1.312	1.1235 1.1230	1.1395 1.1390	1.1545 1.1540	1.1325 1.1315	1.1485 1.1475	1.1635 1.1625	1.124	1.140	1.155
	2	-27 THRU -70					.031		.900 .890				1.1235 1.1230	1.1395 1.1390	1.1545 1.1540	1.1325 1.1315	1.1485 1.1475	1.1635 1.1625	1.124	1.140	1.155
	3	-71 THRU -112					.038		.900 .890				1.1865 1.1860	1.2015 1.2010	1.2175 1.2170	1.1955 1.1945	1.2105 1.2095	1.2265 1.2255	1.187	1.202	1.218

NOTES:

(a) "D" DIA. SHALL BE VERIFIED WITHIN ITS GRIP BEARING PORTION BEFORE EXPANSION; WHEN SLEEVE IS PUSHED TO PIN HEAD, THE "D" DIA. SHALL BE WITHIN THE EXPANDED "D" DIA. LIMITS.

(b) THIS BOLT IS MANUFACTURED AS A PRECISION ASSEMBLY. REMOVAL OF SLEEVE FROM PIN BY USER WILL RESULT IN LOSS OF THE BOLT'S ABILITY TO MEET MIN/MAX EXPANSION LIMITS.

(c) ALL DIMENSIONS ARE IN INCHES. TOLERANCES: DECIMALS $\pm$.010, ANGLES $\pm 2^{\circ}$.

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TABLE II. Mechanical properties.

Dia Dash Number	Ultimate Tensile Strength Lbs-Min (b)	Thread Area In^2 (b)	Double Shear Strength Lbs-Min (a)	Pin Test Diameter (+.0005) (a) In.	Pin Test Dia Area (a) In^2	Max Install Torque In.-Lbs	Fatigue Test Loading Lbs ± 2% (c)	
							Max	Min
-3	3,470	.0217	5,380	.1900	.0284	60	1,600	160
-4	6,210	.0388	9,300	.2500	.0491	115	2,860	285
-5	9,820	.0614	14,600	.3125	.0767	185	4,520	450
-6	15,200	.0950	21,000	.3750	.1104	335	6,990	700
-7	20,600	.1288	28,600	.4375	.1503	540	9,480	950
-8	27,500	.1717	37,300	.5000	.1963	805	12,600	1,260
-9	34,800	.2176	47,200	.5625	.2485	1080	16,000	1,600
-10	43,600	.2724	58,300	.6250	.3068	1560	20,100	2,010
-12	63,200	.3952	83,900	.7500	.4418	2930	29,100	2,910
-14	86,300	.5392	114,200	.8750	.6013	4340	39,700	3,970
-16	112,000	.7027	149,200	1.000	.7854	6100	51,500	5,150

Notes:

a. Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum shear strength of 95 KSI and the stress area at the test diameter.

b. The ultimate tensile strength values are calculated on the basic pitch diameter of the external thread and a tensile strength of 160 KSI.

c. Maximum fatigue load is based on 46% of the ultimate tensile strength. Minimum fatigue load is 10% of maximum fatigue load.

10. Example of military part number.

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- 34 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- \- — '-' designates basic shank diameter: X designates 1st oversize, Y designates 2nd oversize. (Ink stamp on sleeve when applicable.)
- 12 — Diameter dash numbers in sixteenths of an inch.
- M85667/9 — Basic specification sheet number.

11. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.

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TABLE III. Bolt grips and lengths.

Grip Number	Bolt Design Grip	-3 Diameter		-4 Diameter		-5 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
5	.312	.353	.696	.353	.759	.420	.904
6	.375	.415	.758	.415	.821	.483	.967
7	.438	.478	.821	.478	.884	.545	1.029
8	.500	.540	.883	.540	.946	.608	1.092
9	.562	.603	.946	.603	1.009	.670	1.154
10	.625	.665	1.008	.665	1.071	.733	1.217
11	.688	.728	1.071	.728	1.134	.795	1.279
12	.750	.790	1.133	.790	1.196	.858	1.342
13	.812	.853	1.196	.853	1.259	.920	1.404
14	.875	.915	1.258	.915	1.321	.983	1.467
15	.938	.978	1.321	.978	1.384	1.045	1.529
16	1.000	1.040	1.383	1.040	1.446	1.108	1.592
17	1.062	1.103	1.446	1.103	1.509	1.170	1.654
18	1.125	1.165	1.508	1.165	1.571	1.233	1.717
19	1.188	1.228	1.571	1.228	1.634	1.295	1.779
20	1.250	1.290	1.633	1.290	1.696	1.358	1.842
21	1.312	1.353	1.696	1.353	1.759	1.420	1.904
22	1.375	1.415	1.758	1.415	1.821	1.483	1.967
23	1.438				1.884	1.545	2.029
24	1.500				1.946	1.608	2.092
25	1.562				2.009	1.670	2.154
26	1.625				2.071	1.733	2.217
27	1.688				2.134	1.795	2.279
28	1.750				2.196	1.858	2.342
29	1.812					1.920	2.404
30	1.875					1.983	2.467

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-6 Diameter		-7 Diameter		-8 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
6	.375	.420	.982	.425	1.050	.425	1.128
7	.438	.483	1.045	.488	1.113	.488	1.191
8	.500	.545	1.107	.550	1.175	.550	1.253
9	.562	.608	1.170	.613	1.238	.613	1.316
10	.625	.670	1.232	.675	1.300	.675	1.378
11	.688	.733	1.295	.738	1.363	.738	1.441
12	.750	.795	1.357	.800	1.425	.800	1.503
13	.812	.858	1.420	.863	1.488	.863	1.566
14	.875	.920	1.482	.925	1.550	.925	1.628
15	.938	.983	1.545	.988	1.613	.988	1.691
16	1.000	1.045	1.607	1.050	1.675	1.050	1.753
17	1.062	1.108	1.670	1.113	1.738	1.113	1.816
18	1.125	1.170	1.732	1.175	1.180	1.175	1.878
19	1.188	1.233	1.795	1.238	1.863	1.238	1.941
20	1.250	1.295	1.857	1.300	1.925	1.300	2.003
21	1.312	1.358	1.920	1.363	1.988	1.363	2.066
22	1.375	1.420	1.982	1.425	2.050	1.425	2.128
23	1.438	1.483	1.483	1.488	2.113	1.488	2.191
24	1.500	1.545	2.107	1.550	2.175	1.550	2.253
25	1.562	1.608	2.170	1.613	2.238	1.613	2.316
26	1.625	1.670	2.232	1.675	2.300	1.675	2.378
27	1.688	1.733	2.295	1.738	2.363	1.738	2.441
28	1.750	1.795	2.357	1.800	2.425	1.800	2.503
29	1.812	1.858	2.420	1.863	2.488	1.863	2.566
30	1.875	1.920	2.482	1.925	2.550	1.925	2.628
31	1.938	1.983	2.545	1.988	2.613	1.988	2.691
32	2.000	2.045	2.607	2.050	2.675	2.050	2.753
33	2.062	2.108	2.670	2.113	2.738	2.113	2.816
34	2.125	2.170	2.732	2.175	2.800	2.175	2.878
35	2.188			2.238	2.863	2.238	2.941
36	2.250			2.300	2.925	2.300	3.003
37	2.312			2.363	2.988	2.363	3.066
38	2.375			2.425	3.050	2.425	3.128
39	2.438			2.488	3.113	2.488	3.191
40	2.500			2.550	3.175	2.550	3.253
41	2.562					2.613	3.316
42	2.625					2.675	3.378
43	2.688					2.738	3.441
44	2.750					2.800	3.503
45	2.812					2.863	3.566
46	2.875					2.925	3.628

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-9 Diameter		-10 Diameter		-12 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
10	.625	.680	1.461	.680	1.523	.685	1.685
11	.688	.743	1.524	.743	1.586	.748	1.748
12	.750	.805	1.586	.805	1.648	.810	1.810
13	.812	.868	1.649	.868	1.711	.873	1.873
14	.875	.930	1.711	.930	1.773	.935	1.935
15	.938	.993	1.774	.993	1.836	.998	1.998
16	1.000	1.055	1.836	1.055	1.898	1.060	2.060
17	1.062	1.118	1.899	1.118	1.961	1.123	2.123
18	1.125	1.180	1.961	1.180	2.023	1.185	2.185
19	1.188	1.243	2.024	1.243	2.086	1.248	2.248
20	1.250	1.305	2.086	1.305	2.148	1.310	2.310
21	1.312	1.368	2.149	1.368	2.211	1.373	2.373
22	1.375	1.430	2.211	1.430	2.273	1.435	2.435
23	1.438	1.493	2.274	1.493	2.336	1.498	2.498
24	1.500	1.555	2.336	1.555	2.398	1.560	2.560
25	1.562	1.618	2.399	1.618	2.461	1.623	2.623
26	1.625	1.680	2.461	1.680	2.523	1.685	2.685
27	1.688	1.743	2.524	1.743	2.586	1.748	2.748
28	1.750	1.805	2.586	1.805	2.648	1.810	2.810
29	1.812	1.868	2.649	1.868	2.711	1.873	2.873
30	1.875	1.930	2.711	1.930	2.773	1.935	2.935
31	1.938	1.993	2.774	1.993	2.836	1.998	2.998
32	2.000	2.055	2.836	2.055	2.898	2.060	3.060
33	2.062	2.118	2.899	2.118	2.961	2.123	3.123
34	2.125	2.180	2.961	2.180	3.023	2.185	3.185
35	2.188	2.243	3.024	2.243	3.086	2.248	3.248
36	2.250	2.305	3.086	2.305	3.148	2.310	3.310
37	2.312	2.368	3.149	2.368	3.211	2.373	3.373
38	2.375	2.430	3.211	2.430	3.273	2.435	3.435
39	2.438	2.493	3.274	2.493	3.336	2.498	3.498
40	2.500	2.555	3.336	2.555	3.398	2.560	3.560
41	2.562	2.618	3.399	2.618	3.461	2.623	3.623
42	2.625	2.680	3.461	2.680	3.523	2.685	3.685
43	2.688	2.743	3.524	2.743	3.586	2.748	3.748
44	2.750	2.805	3.586	2.805	3.648	2.810	3.810
45	2.812	2.868	3.649	2.868	3.711	2.873	3.873
46	2.875	2.930	3.711	2.930	3.773	2.935	3.935
47	2.938	2.993	3.774	2.993	3.836	2.998	3.998
48	3.000	3.055	3.836	3.055	3.898	3.060	4.060
49	3.062	3.118	3.899	3.118	3.961	3.123	4.123

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-9 Diameter		-10 Diameter		-12 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
50	3.125	3.180	3.961	3.180	4.023	3.185	4.185
51	3.188	3.243	4.024	3.243	4.086	3.248	4.248
52	3.250	3.305	4.086	3.305	4.148	3.310	4.310
53	3.312	3.368	4.149	3.368	4.211	3.373	4.373
54	3.375	3.430	4.211	3.430	4.273	3.435	4.435
55	3.438	3.493	4.274	3.493	4.336	3.498	4.498
56	3.500	3.555	4.336	3.555	4.398	3.560	4.560
57	3.562	3.618	4.399	3.618	4.461	3.623	4.623
58	3.625	3.680	4.461	3.680	4.523	3.685	4.685
59	3.688	3.743	4.524	3.743	4.586	3.748	4.748
60	3.750	3.805	4.586	3.805	4.648	3.810	4.810
61	3.812	3.868	4.649	3.868	4.711	3.873	4.873
62	3.875	3.930	4.711	3.930	4.773	3.935	4.935
63	3.938	3.993	4.774	3.993	4.836	3.998	4.998
64	4.000	4.055	4.836	4.055	4.898	4.060	5.060
65	4.062	4.118	4.899	4.118	4.961	4.123	5.123
66	4.125	4.180	4.961	4.180	5.023	4.185	5.185
67	4.188	4.243	5.024	4.243	5.086	4.248	5.248
68	4.250	4.305	5.086	4.305	5.148	4.310	5.310
69	4.312	4.368	5.149	4.368	5.211	4.373	5.373
70	4.375	4.430	5.211	4.430	5.273	4.435	5.435
71	4.438	4.493	5.274	4.493	5.336	4.498	5.498
72	4.500	4.555	5.336	4.555	5.398	4.560	5.560
73	4.562			4.618	5.461	4.623	5.623
74	4.625			4.680	5.523	4.685	5.685
75	4.688			4.743	5.586	4.748	5.748
76	4.750			4.805	5.648	4.810	5.810
77	4.812			4.868	5.711	4.873	5.873
78	4.875			4.930	5.773	4.935	5.935
79	4.938					4.998	5.998
80	5.000					5.060	6.060
81	5.062					5.123	6.123
82	5.125					5.185	6.185
83	5.188					5.248	6.248
84	5.250					5.310	6.310
85	5.312					5.373	6.373
86	5.375					5.435	6.435
87	5.438					5.498	6.498
88	5.500					5.560	6.560
89	5.562					5.623	6.623
90	5.625					5.685	6.685

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-14 Diameter		-16 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
15	.938	.998	2.154	.999	2.311
16	1.000	1.060	2.216	1.062	2.373
17	1.062	1.123	2.279	1.124	2.436
18	1.125	1.185	2.341	1.186	2.498
19	1.188	1.248	2.404	1.249	2.561
20	1.250	1.310	2.466	1.311	2.623
21	1.312	1.373	2.529	1.374	2.686
22	1.375	1.435	2.591	1.436	2.748
23	1.438	1.498	2.654	1.499	2.811
24	1.500	1.560	2.716	1.561	2.873
25	1.562	1.623	2.779	1.624	2.936
26	1.625	1.685	2.841	1.686	2.998
27	1.688	1.748	2.904	1.749	3.061
28	1.750	1.810	2.966	1.811	3.123
29	1.812	1.873	3.029	1.874	3.186
30	1.875	1.935	3.091	1.936	3.248
31	1.938	1.998	3.154	1.999	3.311
32	2.000	2.060	3.216	2.061	3.373
33	2.062	2.123	3.279	2.124	3.436
34	2.125	2.185	3.341	2.186	3.498
35	2.188	2.248	3.404	2.249	3.561
36	2.250	2.310	3.466	2.311	3.623
37	2.312	2.373	3.529	2.374	3.686
38	2.375	2.435	3.591	2.436	3.748
39	2.438	2.498	3.654	2.499	3.811
40	2.500	2.560	3.716	2.561	3.873
41	2.562	2.623	3.779	2.624	3.936
42	2.625	2.685	3.841	2.686	3.998
43	2.688	2.748	3.904	2.749	4.061
44	2.750	2.810	3.966	2.811	4.123
45	2.812	2.873	4.029	2.874	4.186
46	2.875	2.935	4.091	2.936	4.248
47	2.938	2.998	4.154	2.999	4.311
48	3.000	3.060	4.216	3.061	4.373
49	3.062	3.123	4.279	3.124	4.436
50	3.125	3.185	4.341	3.186	4.498
51	3.188	3.248	4.404	3.249	4.561
52	3.250	3.310	4.466	3.311	4.623

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-14 Diameter		-16 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
53	3.312	3.373	4.529	3.374	4.686
54	3.375	3.435	4.591	3.436	4.748
55	3.438	3.498	4.654	3.499	4.811
56	3.500	3.560	4.716	3.561	4.873
57	3.562	3.623	4.779	3.624	4.936
58	3.625	3.685	4.841	3.686	4.998
59	3.688	3.748	4.904	3.749	5.061
60	3.750	3.810	4.966	3.811	5.123
61	3.812	3.873	5.029	3.874	5.186
62	3.875	3.935	5.091	3.936	5.248
63	3.938	3.998	5.154	3.999	5.311
64	4.000	4.060	5.216	4.061	5.373
65	4.062	4.123	5.279	4.124	5.436
66	4.125	4.185	5.341	4.186	5.498
67	4.188	4.248	5.404	4.249	5.561
68	4.250	4.315	5.471	4.311	5.623
69	4.312	4.378	5.534	4.374	5.686
70	4.375	4.440	5.596	4.436	5.748
71	4.438	4.503	5.659	4.506	5.818
72	4.500	4.565	5.721	4.568	5.880
73	4.562	4.628	5.784	4.631	5.943
74	4.625	4.690	5.846	4.693	6.005
75	4.688	4.753	5.909	4.756	6.068
76	4.750	4.815	5.971	4.818	6.130
77	4.812	4.878	6.034	4.881	6.193
78	4.875	4.940	6.096	4.943	6.255
79	4.938	5.003	6.159	5.006	6.318
80	5.000	5.065	6.221	5.068	6.380
81	5.062	5.128	6.284	5.131	6.443
82	5.125	5.190	6.346	5.193	6.505
83	5.188	5.253	6.409	5.256	6.568
84	5.250	5.315	6.471	5.318	6.630
85	5.312	5.378	6.534	5.381	6.693
86	5.375	5.440	6.596	5.443	6.755
87	5.438	5.503	6.659	5.506	6.818
88	5.500	5.565	6.721	5.568	6.880
89	5.562	5.628	6.784	5.631	6.943
90	5.625	5.690	6.846	5.693	7.005

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-14 Diameter		-16 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
91	5.688	5.753	6.909	5.756	7.068
92	5.750	5.815	6.971	5.818	7.130
93	5.812	5.878	7.034	5.881	7.193
94	5.875	5.940	7.096	5.943	7.255
95	5.938	6.003	7.159	6.006	7.318
96	6.000	6.065	7.221	6.068	7.380
97	6.062	6.128	7.284	6.131	7.443
98	6.125	6.190	7.346	6.193	7.505
99	6.188	6.253	7.409	6.256	7.568
100	6.250	6.315	7.471	6.318	7.630
101	6.312	6.378	7.534	6.381	7.693
102	6.375	6.440	7.596	6.443	7.755
103	6.438	6.503	7.659	6.506	7.818
104	6.500	6.565	7.721	6.658	7.880
105	6.562	6.628	7.784	6.631	7.943
106	6.625	6.690	7.846	6.693	8.005
107	6.688	6.753	7.909	6.756	8.068
108	6.750	6.815	7.971	6.818	8.130
109	6.812			6.881	8.193
110	6.875			6.943	8.255
111	6.938			7.006	8.318
112	7.000			7.068	8.380
113	7.062			7.131	8.443
114	7.125			7.193	8.505
115	7.188			7.256	8.568
116	7.250			7.318	8.630
117	7.312			7.381	8.693
118	7.375			7.443	8.755
119	7.438			7.506	8.818
120	7.500			7.568	8.880
121	7.562			7.631	8.943
122	7.625			7.693	9.005
123	7.688			7.756	9.068
124	7.750			7.818	9.130
125	7.812			7.881	9.193
126	7.875			7.943	9.255
127	7.938			8.006	9.318
128	8.000			8.068	9.380

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b. The bolt assemblies identified herein are proprietary products and are covered by the following patents: U.S. 4,087,896 (expiring May 9, 1997); 4,106,036 (expiring July 25, 1997); 4,048,898 (expiring September 20, 1996); Great Britain 1,524,886; Canada 1,023,585; Japan 50-125256; Italy 1,051,593; Israel 515578; and Germany 2,546,641.5. The Government does not have a royalty-free license.

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Review:
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